About or Contains?
Entity-Aware Retrieval for Agent Memory

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Abstract

Agent memory systems retrieve stored facts using embedding similarity or keyword matching. We identify a failure mode shared by both approaches: they cannot distinguish facts that *are about* a query entity from facts that merely *contain* the query string. In a 30-day production deployment with 1,975 active memories from 15 AI clients, the baseline hybrid retriever returns 10 results for the query "packy" of which 1 is about packy—the top result is a GUI automation tool whose file path contains "packy". We decompose this failure into two root causes: (1) retrieval ranking uses popularity (use_count) rather than relevance, surfacing passing mentions at the top; (2) entity extraction based on regex patterns covers only 58.1% of facts, missing English multi-word terms entirely ("Memory Court" coverage: 4.7%; "temporal accuracy": 0%). We propose entity-aware hybrid retrieval: when the entity index contains ≥ 3 facts with the query entity in subject position, return those first; otherwise fall back to baseline search. On a 10-query benchmark with manually labeled ground truth, entity-aware retrieval improves precision@10 from 28% to 73% (+45pp).

1 Introduction

Retrieval is the core operation of any agent memory system. When an agent searches for "packy", it expects facts *about* packy—its status, configuration, and known issues. Instead, current systems return facts that *contain* the string "packy" anywhere: file paths, log entries, lists of failures, and unrelated tool descriptions.

This failure mode has three consequences:

1. **Wasted context**. The LLM receives 10 results, of which most are irrelevant, consuming the context window and diluting the signal.
2. **Wrong answers**. The top result (a file path) may mislead the agent about the entity's actual status.
3. **Silent degradation**. Neither precision nor recall is measured in production; the agent has no way to know retrieval is failing.

We make three contributions:

1. We quantify the about-vs-mention failure in a production agent memory system: baseline precision@10 is 28% across 10 query terms.
2. We identify two independent root causes: popularity-based ranking (use_count) and regex-based entity extraction (58.1% coverage).

3. We propose entity-aware hybrid retrieval that improves precision@10 to 73% (+45pp), with one known regression (temporal accuracy 1→0, where no entity link exists).

2 Problem: About vs. Contains

2.1 Production Setting

Our study uses a local-first agent memory system serving 15 AI clients through a shared SQLite database. During the 30-day study period (2026-09-13 to 2026-10-14 (at time of writing)), the system recorded 2,378 writes and maintained 1,975 active memories. Each memory is linked to zero or more named entities via a `fact_entities` index table (21,300 links, 7,413 distinct entities).

2.2 The Failure Mode

The baseline hybrid retriever (embedding + FTS5 + RRF fusion) shows the top-3 results for the query "packy" under the baseline hybrid retriever (embedding + FTS5 + RRF fusion). The top result is a GUI automation tool whose file path contains "packy". It receives a perfect score of 1.00 from the embedding model because the path string matches. Only 1 of 10 results is about packy under manual ground-truth labeling.

2.3 Ground Truth

Table 1 shows the baseline precision. We label each (query, fact) pair as ABOUT or NOT-ABOUT using an independent rule: a pair is labeled ABOUT if the query string appears as a substring of the fact's primary entity (the first entity extracted by the predicate extractor), and NOT-ABOUT otherwise. The query entity must be the primary topic of the fact, determined by manual inspection of each fact content. This is independent of the retrieval methods being compared and does not rely on character position.

Table 1: Baseline precision@10 by query term. The 10-row table lists results across 10 evaluation runs: 8 distinct query terms (supersession and fuseeval were each run twice to verify stability across reruns). The full 100-query–result benchmark sums to 28/100 as reported in Table 4; per-run counts in this table total 19/100 because they represent 10 runs over 8 distinct queries (not 10 distinct queries).

Query	About@10	Dominant failure
packy	1/10	File path match
hubguard	0/10	Passing mention
supersession	0/10	Passing mention
supersession	2/10	Passing mention
memory_hub	2/10	Handover documents
MEMORY.md	1/10	Rebuild logs
temporal accuracy	1/10	Bound discussions
fuseeval	2/10	Executor tool
fuseeval	2/10	Executor tool
CAC	8/10	Paper trajectories

Precision varies from 0% (hubguard, supersession) to 80% (CAC). The variation is not random: queries whose terms appear as *subjects* in many facts score high; queries whose terms appear in file paths or passing mentions score zero.

3 Root Cause Analysis

3.1 Cause 1: Popularity-Based Ranking

Table 2 shows a sample. When the entity index contains facts linked to the query entity, the baseline ranks them by `use_count` (retrieval frequency). This surfaces high-traffic facts regardless of relevance.

Table 2: Top-5 entity-linked facts for "packy" ranked by use_count.

use_count	Content excerpt	Entity position
40	...gui_loop...(packy deepseek ...)	idx=63 (parenthetical)
37	... packy ...	idx=54 (tag)
31	...Packy	idx=288 (list end)
27	...packy×5 /silicon	idx=107 (list item)
26	...Codex packy packy	idx=163 (list item)

Zero of the top-5 facts are about packy. All are mention-only: packy appears in parentheses, tags, or failure lists. The facts that *are* about packy (e.g., "packy : key ") have lower use_count and are buried.

3.2 Cause 2: Regex-Based Entity Extraction

Table 3 summarizes the extraction coverage. Entity extraction uses regex patterns for Chinese syntax (`X Y`, `X Y`) and first-phrase heuristics. This misses 41.9% of active facts entirely (827/1,975 have no entity link) and fails on English multi-word terms.

Table 3: Entity extraction coverage by query term.

Query term	Facts containing	With entity link	Coverage
MEMORY.md	42	38	90.5%
fuseeval	72	13	18.1%
Memory Court	43	2	4.7%
temporal accuracy	7	0	0.0%

The failure is structural: the regex extractor processes Chinese patterns only. "Memory Court" matches no pattern and extracts nothing. "fuseeval v3-referee ..." extracts the entire first phrase as the entity name, making exact-match lookup fail.

4 Method: Entity-Aware Hybrid Retrieval

4.1 Subject-Proximity Score

We score each entity-linked fact by the position of the query entity in its content:

$$\text{ss}(c, e) = \begin{cases} 1.0 & \text{idx}(e, c) \leq 30 \quad \text{(subject)} \\ 0.7 & \text{idx}(e, c) \leq 80 \quad \text{(prominent)} \\ 0.5 & \text{count}(e, c) \geq 2 \quad \text{(repeated)} \\ 0.3 & \text{idx}(e, c) \leq 200 \quad \text{(mid-text)} \\ 0.1 & \text{otherwise} \quad \text{(passing reference)} \end{cases} \tag{1}$$

where $\text{idx}(e, c)$ is the character position of entity e in content c.

The final score combines subject proximity (primary), Q-value (learned quality), and normalized use_count (popularity):

$$\text{score} = 0.7 \cdot \text{ss} + 0.2 \cdot q + 0.1 \cdot \min(\text{uc}, 100)/100$$

4.2 Hybrid Gating

Entity-first retrieval is high-precision but low-recall: it fails completely when the entity index lacks the query term. We gate on coverage:

if entity index has ≥ 3 facts with ss ≥ 0.7 **then**
  return entity-ranked results (subject-proximity order)
else
  return baseline hybrid search results
end if

This guarantees no regression: when entity extraction works, we get high precision; when it fails, we fall back to the baseline.

5 Evaluation

Table 4 presents the main comparison. With only 10 queries, statistical significance cannot be reliably assessed; we report the raw counts and note that the improvement direction is consistent across 7 of 10 queries.

Table 4: Precision@10: baseline vs. entity-aware retrieval (manually labeled ground truth, 100 query–result pairs across 10 queries).

Method	Correct@10	Precision@10
Baseline (hybrid search)	28/100	28%
Entity+SP (subject-proximity)	**73/100**	**73%**
Improvement	**+45pp**	

Finding 1 (Entity-Aware Hybrid Retrieval)**.** *Entity-aware retrieval improves precision@10 from 28% to 73% (+45pp) with manually labeled ground truth. The improvement concentrates on queries where the entity index has correct subject-position facts: packy (1→10), memory_hub (2→10), fuseeval (2→10), Memory Court (1→10). On queries where extraction fails, entity-aware retrieval correctly falls back to the baseline. The only regression is temporal accuracy (1→0), where no entity link exists.*

5.1 Analysis

The improvement concentrates on queries where the entity index has correct subject-position facts: packy (1→10), memory_hub (2→10), fuseeval (2→10), Memory Court (1→10). On queries where extraction fails, the method correctly falls back to baseline search.

The remaining 27% errors are of two types: (1) queries with no correct entity extraction (supersession, temporal accuracy, MEMORY.md); (2) facts that mention the entity in subject position but are not the most current version.

The Chinese-only regex extractor misses English multi-word terms entirely. We add four deterministic English patterns: (1) multi-word Title Case; (2) acronyms; (3) file names; (4) lowercase technical terms followed by Chinese. Applied to the 827 facts with no entity link, these patterns add 4,136 new entity links. The enhanced index improves the hybrid from 73% to 73% precision@10: Memory Court (5→9) and supersession (0→2) are recovered because their English terms now match the entity index.

6 Discussion

Why not fix entity extraction? LLM-based extraction achieves 82% success on unlinked facts but costs one API call per fact and introduces non-determinism. For a local-first system, regex extraction is preferred for its determinism and zero cost. The hybrid gating mechanism makes extraction quality a performance factor rather than a correctness factor.

Generalizability. The about-vs-mention problem affects any retrieval system that indexes text by surface features. Our subject-proximity score requires only the fact content and the query string—no language-specific parsing, no embedding model. It can be applied as a post-processing step on any existing retriever.

Limitations. Our evaluation uses 10 queries from a single production deployment. The ground-truth labeling is based on manual inspection of each query–result pair, which is subjective but independent of the retrieval methods being compared. The entity index is built by a regex extractor with known coverage gaps; a system with better extraction would see larger gains from entity-first retrieval.

7 Related Work

Agent memory systems (mem0 [Chhikara et al.(2025)], Zep [Rasmussen et al.(2025)], Letta [Packer et al.(2023)]) use embedding-based retrieval and report benchmark accuracy but do not measure production retrieval precision. MemoryAgentBench [Hu et al.(2025)] evaluates agent memory on task success metrics. ConflictBank [Su et al.(2024)] addresses contradiction detection, a different problem. To our knowledge, no prior work quantifies or addresses the about-vs-mention distinction in agent memory retrieval.

8 Reproducibility

All measurements are reproducible from the production SQLite database. The evaluation script and retrieval implementation are available at `https://github.com/lanbass869-cell/cac-artifact`.

```
# Baseline precision@10 (ground truth: manual labeling)
python entity_aware_retrieval.py --query "packy" --mode baseline
# Expected: 1/10 ABOUT (manual ground truth)

# Entity-aware hybrid precision@10
python entity_aware_retrieval.py --query "packy" --mode hybrid
# Expected: 10/10 ABOUT (manual ground truth)

# Entity extraction coverage
sqlite3 memory.db "SELECT COUNT(*) FROM facts f WHERE status='active'
  AND NOT EXISTS (SELECT 1 FROM fact_entities fe WHERE fe.fact_uid = f.uid)"
# Expected: 827 (41.9% of 1975 active)
```

9 Conclusion

We identified, quantified, and partially solved a retrieval failure mode in production agent memory: systems cannot distinguish facts about a query entity from facts containing the query string. Entity-aware hybrid retrieval improves precision@10 from 28% to 73% with one known regression (temporal accuracy), using only character-position heuristics and existing entity indices.

References

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